Characterisation of ordinals: a set is an ordinal exactly when it is a transitive set of ordinals

The Open Logic Project contributors

openlogic:set-theory/ordinals:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-set-theory-ordinals/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: both directions with their three named internal citations appear verbatim in the span; the forward direction is explicitly labelled Left-to-right by the author.

Editorial note. The label corordtransitiveord names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own cor environment in the unit source.

Allowed standard prerequisites (editorial).

- Elements of ordinals are ordinals. Same unit, \ollabel{ordmemberord} (source line 416). Left-to-right direction (line 539): an ordinal's transitivity and element-ordinality give the forward inclusion.
- Transfinite induction schema. Same unit, \ollabel{ordinductionschema} (source line 446). Right-to-left (line 541): one of the two inputs showing membership well-orders the transitive set A.
- Trichotomy for ordinal membership. Same unit, \ollabel{ordtrichotomy} (source line 477). Right-to-left (line 541): the second input, giving the linear-order half of the well-order claim.

Source reference. Original reference ordinductionschema: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 446–446. Prerequisite pointer invoked at proof line(s) 541; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference ordmemberord: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 416–416. Prerequisite pointer invoked at proof line(s) 539; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference ordtrichotomy: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 477–477. Prerequisite pointer invoked at proof line(s) 541; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Corollary 0.1. *A is an ordinal iff A is a transitive set of ordinals.*

Proof. Left-to-right. By ordmemberord. *Right-to-left.* If A is a transitive set of ordinals, then $\in$ well-orders A by ordinductionschema and ordtrichotomy. □